

## 1 What is face-api.js?

**face-api.js** is a JavaScript library that uses **deep learning (CNN)** to:

- Detect faces
- Identify whose face it is
- Recognize expressions
- Find face landmarks (eyes, nose, mouth)

It runs in the browser using **TensorFlow.js**.

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## 2 What is CNN? (Convolutional Neural Network)

A **CNN (Convolutional Neural Network)** is a type of Artificial Intelligence model designed to understand images.

Think of it like this:

- 👉 A normal computer sees an image as numbers (pixels).
- 👉 A CNN learns patterns inside those pixels.

For example:

When the system sees a face, it does NOT think:

“Oh that’s John.”

Instead, it thinks like this:

- I see two dark areas (eyes)
- I see a vertical shape (nose)
- I see a curved line (mouth)
- I see distances between features

Then it converts that into a **face pattern number**.

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## 3 How Your Face Recognition Attendance Works

Your system likely works like this:

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### Step 1: Face Detection

Model used:

- `tiny_face_detector`
- or `ssd_mobilenetv1`

What happens:

The camera scans the image and asks:

“Is there a face here?”

It draws a box around the face.

This is the FIRST CNN model.

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## **Step 2: Face Landmarks**

Model used:

- `face_landmark_68_model`
- or `face_landmark_68_tiny_model`

This finds:

- 68 points on the face
- Eyes
- Nose
- Mouth
- Jawline

Why?

Because face recognition is about **distances and structure**.

Example:

- Distance between eyes
  - Distance from nose to mouth
  - Shape of jaw
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## **Step 3: Face Recognition Model**

Model used:

- `face_recognition_model`

This is the most important one.

It converts a face into a:

👉 **128-number vector (face descriptor)**

Example (simplified):

Instead of:

This is John

It becomes:

[0.23, -0.11, 0.78, 0.09, ... 128 numbers]

That number set is like a **face fingerprint**.

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#### **Step 4: Matching Faces**

When someone comes to church:

1. Camera captures face
2. System converts it into 128 numbers
3. It compares those numbers with saved members

If the numbers are very close →

✅ Mark attendance

If not close →

❌ Unknown person

This is called **Euclidean distance comparison**.

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#### **👉 Why It's Called CNN**

CNN works in layers:

##### **Layer 1 – Detect Edges**

Finds lines and edges in the image.

##### **Layer 2 – Detect Shapes**

Combines edges into shapes like eyes and nose.

##### **Layer 3 – Detect Face Pattern**

Combines shapes into full face pattern.

##### **Final Layer – Convert to Numbers**

Creates the 128-number face descriptor.

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## **5 Why CNN is Good for Faces**

Because faces are:

- Complex
- Slightly different every time
- Affected by lighting
- Affected by angle

CNN can learn patterns even if:

- Person wears glasses
  - Hair changes
  - Lighting changes slightly
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## **6 Models You Have in Screenshot**

From your screenshot, I see:

- `age_gender_model` → predicts age & gender
- `face_expression_model` → detects emotions
- `face_landmark_68_model`
- `face_recognition_model`
- `ssd_mobilenetv1_model`
- `tiny_face_detector_model`

That means your system can:

- ✓ Detect face
- ✓ Recognize member
- ✓ Predict age
- ✓ Detect emotions

That's actually advanced.

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## **7 Simple Real-Life Analogy**

Imagine this:

You meet 500 church members.

You don't remember their names.

But your brain remembers:

- Shape of their face
- Eye spacing
- Smile shape

So when they come again, your brain says:

"I've seen this face before."

Your CNN system is doing the same thing — but using math.

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## **8 Important Concept: Training vs Pre-trained**

You did NOT train the CNN from scratch.

face-api.js uses:

### **👉 Pre-trained models**

That means:

- Someone trained it on thousands/millions of faces
- You are just using it for recognition

Your system only stores:

- Face descriptors of your members